

AI Preparedness Index: Ghana, Africa, and the World

Background

AI preparedness Index (AIPI) assesses how ready a country is to harness the benefits and manage the risks of Artificial Intelligence. It is built on four macro-structural pillars: Digital Infrastructure, Human Capital, Innovation, and Regulation. AIPI simply measures a country's readiness for the AI era. It doesn't just look at how many technologies a country has; it looks at the whole ecosystem required for AI to thrive:

- **Digital Infrastructure:** The physical infrastructure such as internet speed, server availability, and connectivity.
- **Human Capital:** The skills of the people, how well the workforce is educated in digital and technical fields.
- **Innovation:** How well AI is being integrated into businesses and the local economy.
- **Regulation:** The laws and ethical frameworks that ensure AI is used safely and effectively.

Problem Statement

There is a significant "AI Divide" globally. While advanced economies are rapidly integrating AI into their infrastructure and labor markets, developing nations, particularly in Africa face foundational hurdles. This project examines where Ghana stands on AI readiness, what's holding it back, and what it would take to close the gap with leaders like Singapore.

Ghana's Position

Ghana's overall AI Preparedness Index (AIPI) is **0.4252**, or **42.52%**. This places Ghana at a global rank of 99th out of 174 countries.

Pillar Analysis for Ghana:

Ghana's Pillar Scores DataFrame:

	Pillar	Index	Percentage
0	Regulation and Ethics	0.133843	13.38%
1	Human Capital and Labor Market Policies	0.113733	11.37%
2	Innovation and Economic Integration	0.088968	8.90%
3	Digital Infrastructure	0.088655	8.87%

- **Strongest Pillar:** Regulation and Ethics (0.1338). This indicates Ghana has a relatively strong framework in terms of AI governance, policies, and ethical considerations. It indicates that Ghana has a presence of policy frameworks or ethical guidelines that provide a stable environment for future AI development.
- **Weakest Pillar:** Digital Infrastructure (0.0887). This is the area with the most significant gap, highlighting a critical need for investment in high-speed internet and reliable data centers. Because AI requires massive amounts of data processing and reliable high-speed connectivity, without the 'pipes' (infrastructure), the 'software' (AI) cannot run effectively. Hence the need for significant investment.

Ghana - Africa Comparison

Ghana's Rank in Africa: 9 out of 48

Country	AIPI	AIPI %	Rank
Seychelles	0.530653	53.07%	1
Mauritius	0.525113	52.51%	2
South Africa	0.496765	49.68%	3
Tunisia	0.465410	46.54%	4
Kenya	0.445192	44.52%	5
Rwanda	0.437400	43.74%	6
Cabo Verde	0.432587	43.26%	7
Morocco	0.429111	42.91%	8
Ghana	0.425200	42.52%	9
Namibia	0.419612	41.96%	10

Within Africa, Ghana ranks **9th out of 48** African nations included in the dataset. This positions Ghana as one of the leading countries on the continent in terms of AI preparedness and therefore positions itself as a regional hub for digital policy

Global Comparison

When compared to the global top 10 leaders, Ghana faces a substantial gap. The average AIPI score for the top 10 countries is significantly higher than Ghana's. For instance, Singapore, the global leader, has an AIPI of 0.8006, nearly double that of Ghana.

Ghana vs. Global Top 10 Average (Pillar-wise):

- **Regulation and Ethics:** Ghana's strongest pillar (0.134) is still considerably lower than the global leaders' average (0.217).
- **Digital Infrastructure:** This gap is most pronounced here, with Ghana's 0.089 lagging far behind the leaders' 0.193.
- **Human Capital:** Ghana's 0.114 indicates a need for continued investment in digital literacy and specialized AI skills, compared to the leaders' 0.180.
- **Innovation and Economic Integration:** At 0.089, Ghana scores almost half of the global average (0.175), reflecting challenges in integrating AI into the local business ecosystem and fostering innovation.

The Solution

To bridge the gap and prepare for an AI-driven future, the following areas are identified:

1. **Infrastructure Investment:** Expanding access to high-speed internet and developing robust local data centers to support AI processing.
2. **Upskilling the Labor Market:** Moving beyond basic digital literacy to specialized AI skills and education to bridge the gap in human capital through vocational training.
3. **Fostering Local Innovation:** Encouraging the integration of AI into local businesses and supporting tech startups to move the needle on Innovation and Economic Integration. Moreover, the government should focus on creating an environment that encourages AI startups, integrates AI into existing industries, and promotes research and development in AI.
4. **Leveraging Regulation:** Using the existing strong regulatory framework to build trust and attract international investment in safe, ethical AI projects.

Summary of Findings

While Ghana has built a solid regulatory foundation and has a relatively skilled workforce compared to many other African nations, the lack of digital infrastructure and innovation are the primary factors holding back its global AI competitiveness. Moreover, while AI boost productivity and growth it has its risks. Advanced economies are more vulnerable to job displacement, whereas low-income countries like Ghana face fewer immediate disruptions but are at risk of falling further behind due to lacking infrastructure.

Resources

IMF, AI Preparedness Index

<https://www.imf.org/external/datamapper/datasets/AIPI>

IMF AI, Preparedness Documentation:

<https://www.imf.org/external/datamapper/AIPINote.pdf>

Gen AI, Artificial Intelligence and the Future of Work

<https://www.imf.org/en/Publications/Staff-Discussion-Notes/Issues/2024/01/14/Gen-AI-Artificial-Intelligence-and-the-Future-of-Work-542379>

Tools:

Python, Pandas, Seaborn, Matplotlib